

Multiplicity-Inflected Prime Theory (MIPT)

0. Core intuition

A **prime** p is a *place* where a structure is interrogated (localization / fiber / valuation). A **multiplicity** is the *thickness* of what survives at that place (length, intersection number, valuation exponent, dimension of an eigenspace, order of vanishing).

The theory's goal is to turn this intuition into a **portable machine**:

$$\text{\textit{object over }} \mathrm{Spec} \mathbb{Z}, \text{\textit{phenomenon }} \Phi \mapsto \text{\textit{prime-weight function }} m_\Phi(p) \mapsto \text{\textit{weighted prime measure }} \mu_\Phi.$$

1. Prime-Weight Machine (PWM)

1.1 Inputs

- **Base:** $S = \mathrm{Spec} \mathbb{Z}$ (or $\mathrm{Spec} \mathcal{O}_K$).
- **Object:** a scheme $f : X \rightarrow S$, a cycle, a sheaf/representation, a family $\{X_t\}$, etc.
- **Phenomenon Φ :** intersection failure, singularity thickness, ramification, torsion, degeneration, gap structure, etc.

1.2 Output (Level 1)

A **prime weight**

$$m_\Phi : \{p\} \rightarrow \mathbb{R}_{\geq 0}$$

and the induced **prime-weight measure**

$$\mu_\Phi := \sum_p m_\Phi(p) \delta_p.$$

1.3 Strength levels

- **Level 1 (Measure-only):** tail, clustering, correlations of μ_Φ .
 - **Level 2 (Dirichlet/transform):** define
$$\Pi_\Phi(x) = \sum_{p \leq x} m_\Phi(p), \quad \Theta_\Phi(x) = \sum_{p \leq x} m_\Phi(p) \log p,$$
$$F_\Phi(s) = \sum_p \frac{m_\Phi(p) \log p}{p^s} \quad (\Re(s) > 1).$$
 Optional formal Euler product
$$\zeta_\Phi(s) = \prod_p (1 - p^{-s})^{-m(p)}$$
 (formal unless promoted to Level 3).
 - **Level 3 (Explicit-formula capable):** m_Φ arises from genuine Euler factors (ℓ -adic sheaves/motives/automorphic objects), so prime sums link to zeros via explicit formulas.
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2. Canonical constructions (portable templates)

2.1 Prime Intersection Spectrum (PIS)

Setup: proper $f : X \rightarrow S$, cycles Y, Z meeting properly (or via refined intersection theory).

Pushforward divisor formulation (canonical):

$$D_{\{Y,Z\}} = f_*(Y \cdot Z) \in \mathrm{Div}(S) = \bigoplus_p \mathbb{Z} \cdot [p], \quad D_{\{Y,Z\}} = \sum_p M_p(Y,Z) \cdot [p].$$

Define the **PIS coefficient function**

$$\mathrm{PIS}(Y,Z): p \mapsto M_p(Y,Z).$$

Interpretation: M_p is the total intersection thickness contributed by the fiber over p .

Clustering notion: primes with unusually large $M_p(Y, Z)$ are *intersection-cluster primes*.

2.2 Discriminant/Resultant weight (fast arithmetic-geometry toy)

For a polynomial $f \in \mathbb{Z}[x]$:

$$m(p) = v_p(\mathrm{Disc}(f)).$$

Then $m(p) > 0$ iff f has a multiple root mod p , and $m(p)$ measures severity.

Family version (recommended): sample a family $\{f_t\}$; study averaged weights $\mathbb{E}_t[v_p(\mathrm{Disc}(f_t))]$ to obtain a genuine distribution across primes.

2.3 Bad reduction / ramification weight

For an elliptic curve $E/\mathbb{Q}$:

$$m(p) = v_p(\Delta_{\min}(E)),$$

capturing singular fiber thickness. More generally use conductor exponents, Swan conductors, or discriminants for families.

2.4 Gap tensor (purely arithmetic multiplicity field)

Let p_n be the n -th prime and gaps $g_n = p_{n+1} - p_n$. For each prime q define

$$T(n,q) = v_q(g_n).$$

This is a **multiplicity tensor** recording prime-power thickness inside prime gaps.

Empirical objects:

- $\mathbb{P}(T(\cdot, q) = a)$ for fixed q
- correlations $\text{Cov}(T(\cdot, q), T(\cdot, r))$
- conditional laws $\mathbb{E}[T(\cdot, q) \mid p_n \bmod q^k]$

2.5 Local-factor tensor (Level 3 doorway)

For a geometric L -function with local factors

$$L_p(s) = \prod_i (1 - \alpha_{p,i} p^{-s})^{-m_{p,i}}$$

package

$$A[p,i] = \alpha_{p,i}, \quad W[p,i] = m_{p,i}.$$

Here multiplicity is intrinsic (eigenspace dimension). Zeros become emergent global structure from (A, W) .

3. Structural principles (what must hold for the theory to bite)

3.1 Family/tower principle (avoid finite support)

Many single objects have only finitely many bad primes (finite support of weights). To obtain true prime-distribution phenomena, work with:

- families $\{X_t\}, \{f_t\}, \{E_t\}$
- towers/levels (e.g., iterates, congruence levels, division fields)
- varying cycles (Y_t, Z_t)

3.2 Invariance / model dependence control

Weights should be canonical or stable under controlled changes:

- minimal discriminants / minimal models
- stable reduction models
- intersection multiplicities defined in a model-invariant way (pushforward cycle method)

3.3 Multiplicity types (keep distinct)

- **Algebraic:** lengths, Hilbert–Samuel, intersection numbers.
- **Arithmetic:** valuations, conductor exponents, ramification indices.
- **Analytic:** orders of vanishing of zeros. Bridging them requires Level 3 structure.

4. Testable conjecture shapes (falsifiable, scalable)

4.1 Small-prime tail dominance (generic families)

Heuristic null model for valuation-type weights:

$$\mathbb{E}[m(p)] \approx \frac{1}{p-1} \iff \sum_{p \leq x} \mathbb{E}[m(p)] \sim \log \log x.$$

Prediction: mass concentrates heavily at $p = 2, 3, 5$.

4.2 Gap tensor geometric law (baseline)

For fixed prime q , baseline prediction (large-scale):

$$\mathbb{P}(v_q(g_n) \geq a) \approx \frac{1}{\varphi(q^a)} = \frac{1}{q^{a-1}(q-1)},$$

$$\mathbb{P}(v_q(g_n) = a) \approx q^{-a} \text{ for } a \geq 1,$$

with $q = 2$ giving $\mathbb{P}(v_2(g_n) = a) \approx 2^{-a}$ for $a \geq 1$ (excluding the initial prime).

4.3 Correlated failure-mode clustering

Given multiple phenomena Φ_i built on the same family, expect positive correlations:

$$\sum_{p \leq x} m_{\Phi_i}(p) m_{\Phi_j}(p) \text{ exceeds the product of marginals (after normalization).}$$

5. Fast validation protocols (hours-to-days)

5.1 Protocol A: Gap tensor (no geometry needed)

1. Generate first N primes (e.g., 10^6).
2. Compute gaps g_n .
3. For $q \in \{2, 3, 5, 7, 11, 13\}$, compute $v_q(g_n)$ by repeated division.
4. Compare empirical $\mathbb{P}(v_q = a)$ to q^{-a} .
5. Compute correlations across q and conditioning on residues.

Quick falsifier: persistent strong deviation from q^{-a} beyond small-prime edge effects.

5.2 Protocol B: Discriminant weights in a family

1. Sample many polynomials f_t of fixed degree d with bounded coefficients.
2. Compute $\text{Disc}(f_t)$.
3. For primes $p \leq x$, estimate $\mathbb{E}_t[v_p(\text{Disc}(f_t))]$.
4. Test $\approx 1/(p-1)$ decay and small-prime dominance.

5.3 Protocol C: Plane-curve PIS (computational intersection spectra)

1. Choose integer plane curves $C_1 : f = 0, C_2 : g = 0$.
 2. For each $p \leq x$, reduce mod p .
 3. Compute total intersection multiplicity (local algebra / Gröbner basis over $\mathbb{F}_p$).
 4. Plot $p \mapsto M_p(C_1, C_2)$; identify cluster primes; compare to discriminant/resultant primes.
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6. Roadmap (how to grow MIPT)

1. **Pick a flagship instantiation** (Gap tensor / Discriminant family / PIS).
 2. **Specify invariance** and a canonical normal form.
 3. **Define null models** (valuation heuristics, residue equidistribution heuristics).
 4. **Prove bounds** (support size in families, expected mass, moment bounds).
 5. **Promote to Level 3** in cases with genuine L -functions (sheaves/motives), enabling explicit-formula comparisons.
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7. Glossary

- **Place / prime:** a localization / fiber / valuation site of interrogation.
 - **Multiplicity:** thickness (length/intersection), exponent (valuation/conductor), or analytic order (zero multiplicity).
 - **Prime-weight measure:** $\mu_\Phi = \sum m_\Phi(p)\delta_p$.
 - **PIS:** prime-indexed intersection multiplicity coefficients from pushforward of $Y \cdot Z$.
 - **Gap tensor:** $T(n, q) = v_q(p_{n+1} - p_n)$, multiplicity field on prime gaps.
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